

Advanced (Habanero)

- Q1.** What does a two-way table show?
- (A) Relationship between two variables
 - (B) Distribution of a single variable
 - (C) Correlation between three variables
 - (D) Mean of a dataset
 - (E) Median of a dataset
- Q2.** In a two-way table, what is the $x \cdot y$ cell?
- (A) Row total
 - (B) Column total
 - (C) Grand total
 - (D) Joint frequency
 - (E) Marginal frequency
- Q3.** Which of the following is a use of two-way tables?
- (A) To find the mode of a dataset
 - (B) To compare the means of two groups
 - (C) To analyze the relationship between two categorical variables
 - (D) To calculate the range of a dataset
 - (E) To find the median of a dataset
- Q4.** What is the relative frequency of a cell in a two-way table?
- (A) $\frac{\text{cell frequency}}{\text{row total}}$
 - (B) $\frac{\text{cell frequency}}{\text{column total}}$
 - (C) $\frac{\text{cell frequency}}{\text{grand total}}$
 - (D) $\frac{\text{row total}}{\text{cell frequency}}$
 - (E) $\frac{\text{column total}}{\text{cell frequency}}$
- Q5.** If a two-way table shows a strong association between two variables, what can be concluded?
- (A) The relationship is causal
 - (B) The relationship is correlational
 - (C) The variables are independent
 - (D) The variables are dependent
 - (E) The relationship is not significant
- Q6.** How do you calculate the marginal frequency of a row in a two-way table?
- (A) Add the frequencies of all cells in the row
 - (B) Subtract the frequencies of all cells in the row
 - (C) Multiply the frequencies of all cells in the row
 - (D) Divide the frequencies of all cells in the row
 - (E) Find the mode of the row
- Q7.** What is the purpose of calculating relative frequencies in a two-way table?
- (A) To compare the frequencies of different cells
 - (B) To analyze the relationship between two variables
 - (C) To find the mode of a dataset
 - (D) To calculate the range of a dataset
 - (E) To find the median of a dataset
- Q8.** If a two-way table has r rows and c columns, how many cells are in the table?
- (A) $r + c$
 - (B) $r \cdot c$
 - (C) $r - c$
 - (D) r/c
 - (E) c/r
- Q5.** If a two-way table shows a strong association between two

Open Ended Questions (FRQ)

- Q9.** Create a two-way table to analyze the relationship between favorite color (red, blue, green) and favorite sport (soccer, basketball, tennis) for a group of 30 students. Calculate the relative frequencies and interpret the results.
- Q10.** A survey of 50 employees found the following relationship between job satisfaction (high, low) and years of experience (0-5, 6-10, 11+):

Chef's Tips

- Ensure students have a calculator and pencil
- Allow 60 minutes to complete the test
- Remind students to show all work and explain their reasoning